Cost-Aware Architecture Decisions

every box on the diagram has a monthly price

Watch the 4-min video

In one line: treat cost as a design dimension like latency and availability: estimate the main drivers (compute, storage, data transfer, GPUs, managed premiums), compare options by **total cost of ownership** including people's time, and state the trade-off out loud: "twice the cost for half the latency in Europe".

1 "Roughly what would that cost?"

In a design interview you propose replicating every database across three regions, serving video from your own servers, and calling a large GPU model on every request. The interviewer asks one question: **roughly what would that cost each month?** Strong engineers can answer to the nearest order of magnitude, and know which choice dominates the bill. You don't need exact prices; you need the drivers, a few anchor numbers, and the arithmetic.

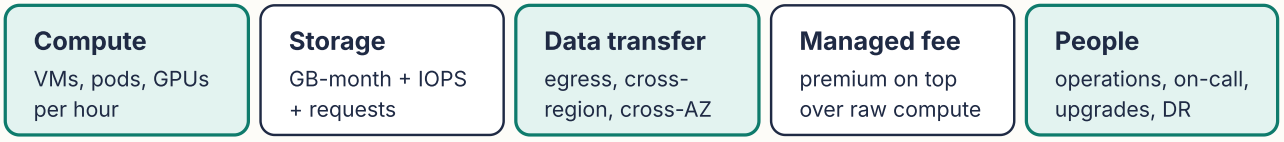


Figure 1. The five cost drivers. The three marked are the ones most often missed in designs.

2 Every decision has a price

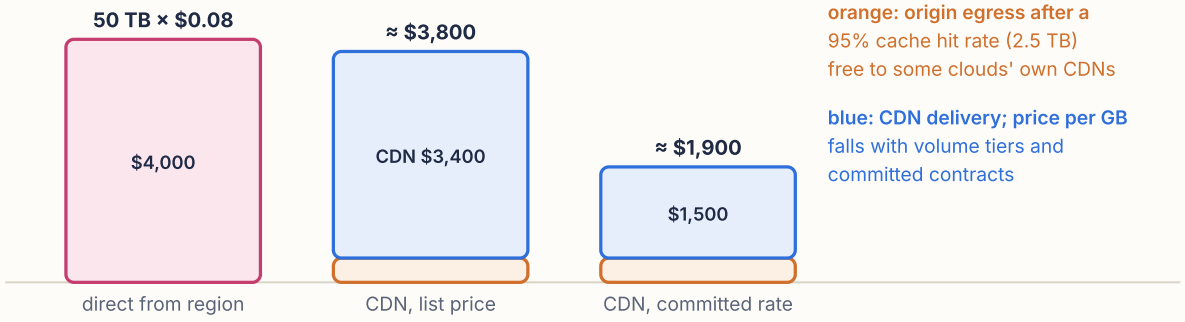
- **Another region** roughly doubles infrastructure for an active-active design, plus cross-region replication traffic. Active-passive with a smaller warm standby costs less, at the price of a slower failover (lesson 52).
- **Synchronous cross-region writes** add latency and inter-region transfer charges on every write.
- **Managed services** charge a premium over raw compute (often 1.5–3× per unit), in exchange for operations you don't do.
- **Higher availability** (lesson 54) usually means more replicas and zones, so more compute and cross-zone traffic.
- **Retention**: keeping every event, log and trace forever is a storage and query cost that grows every month.

3 Data transfer: the classic surprise

Inbound data is usually free; **outbound** is not. Internet egress from the big clouds lists at roughly **$0.05–0.09 per GB**, falling with volume; inter-region transfer is around $0.02 per GB; even traffic between availability zones in one region is charged on some clouds (about $0.01 per GB each way on AWS). A NAT gateway adds a per-GB processing charge (about $0.045 on AWS) on top of egress, so a fleet downloading container images and calling public APIs through NAT pays twice.

WORKED NUMBERS: 50 TB OF VIDEO A MONTH

Served directly from a region at about $0.08/GB: 50,000 GB × $0.08 ≈ **$4,000 a month**. Through a CDN with a 95% cache hit rate, only 2.5 TB leaves the origin (often free or cheap when the CDN belongs to the same cloud), and the CDN's own per-GB price drops with volume tiers and committed contracts. At list prices the saving is modest; with a committed rate or a cheaper CDN, cutting the bill in half or more is realistic, and users get lower latency as a bonus.



Illustrative monthly costs ($ thousands, bar height). Real prices vary by provider, region and contract.

Figure 2. The CDN changes two numbers: how much leaves your origin, and the price per GB for the rest.

4 GPUs dominate AI systems

A single data-centre GPU costs several dollars per hour on demand. At $3 per GPU-hour, one GPU running all year is 3 × 8,760 ≈ **$26,000**; a node with eight of them is over $200,000 a year. So size GPU fleets from first principles:

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GPUs at peak ≈ (peak requests/s × tokens per request) ÷ tokens/s per GPU
                                                      × headroom
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Example with illustrative numbers: 40 requests/s at peak × 500 output tokens = 20,000 tokens/s. If one GPU serving this model with batching sustains about 2,500 tokens/s, you need 8 GPUs, about 10 with headroom: around **$22,000 a month** at $3/hour if they run all the time. The levers that change this most are usually not the hardware: a **smaller model** where it is good enough (often several times cheaper per token), **caching** repeated prompts or answers, **batching** for higher throughput per GPU, shorter prompts and outputs, and autoscaling to the daily curve (lesson 58).

5 Total cost of ownership

The price list is only part of the cost. Self-hosting Kafka on VMs looks cheaper than a managed event service until you add the people: upgrades, security patches, monitoring, capacity planning, on-call, disaster-recovery tests. A fully loaded engineer costs on the order of $10,000–20,000 a month depending on location, so even a third of one is a large line item. Add the **opportunity cost**: those hours aren't spent on the product.

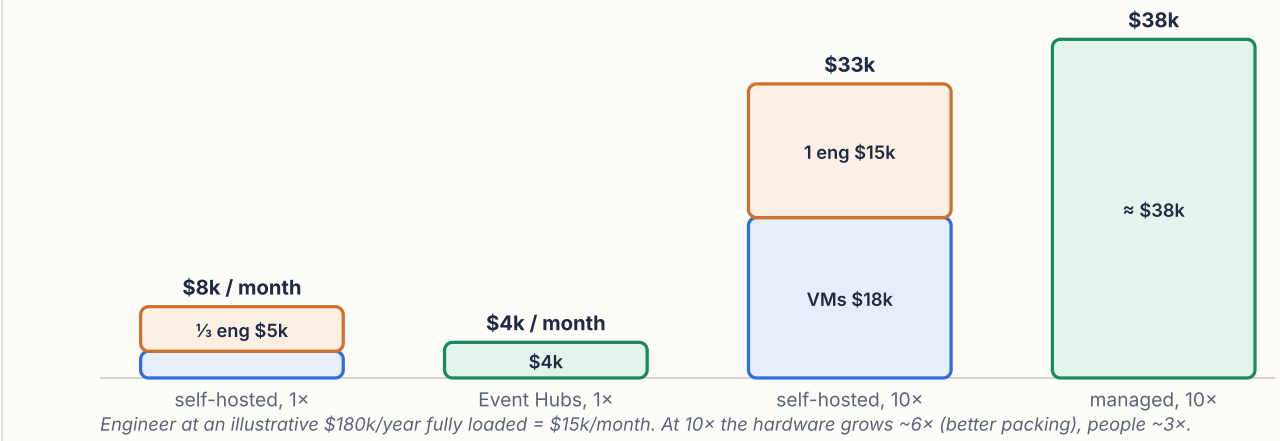


Figure 3. Illustrative TCO for the worked example. Managed wins clearly at 1×; at 10× the gap narrows or reverses, because people scale more slowly than hardware.

At small and medium scale, managed services usually win. At very large scale, self-hosting can pay for itself because the platform team's cost is spread over a big fleet and you can tune hardware precisely. Lock-in, features, compliance and skills also count, but the question to ask is always "**total** cost, at **our** scale, including **people**".

6 Hidden costs to name

Hidden cost	Why it bites	Mitigation
Egress and cross-region transfer	billed per GB, grows with users and replication	CDN, compression, keep replication async and selective
Cross-zone traffic	chatty microservices spread over zones	zone-aware routing; co-locate chatty pairs
NAT gateway and load balancer processing	per-GB and per-hour fees on top of egress	private endpoints to cloud services, fewer LBs (shared ingress)
Observability	logs and metrics ingestion priced per GB	sampling, retention limits, drop debug logs in prod
Storage requests	millions of tiny object PUT/GETs	batch small objects, cache reads
Operational time	self-hosted systems need people	count it in TCO; prefer managed below large scale

7 In the interview

1. Name the **two or three dominant drivers** for this design (GPUs for an AI assistant; egress for video; storage for logging).
2. Estimate them with round numbers you can defend ("egress about 8 cents a GB", "a GPU a few dollars an hour").
3. **Compare two options** and state the trade-off: "this costs about twice as much but halves latency for European users", or "we accept eventual consistency to avoid cross-region write charges".
4. Mention the cheap wins: caching, a CDN, a smaller model, autoscaling, tiering.

WORKED EXAMPLE: SELF-HOSTED KAFKA VS AZURE EVENT HUBS

Self-hosted: six VMs with premium disks, about **$3,000 a month**, plus about a third of an engineer to operate it: upgrades, broker failures, rebalancing, monitoring, on-call. At an illustrative $15,000 a month per engineer, that's $5,000, so **about $8,000** in total. **Event Hubs** at the required throughput costs about **$4,000 a month**, with the operations included and a Kafka-compatible endpoint so clients don't change. For a team of five, the managed option has the lower total cost and frees a third of a person. At ten times the scale, hardware may grow to about $18,000 (better packing) and people to one engineer, about $33,000, while the managed bill might be around $38,000: the answer may flip, and it's worth re-checking with real quotes and a capacity test.

COMMON MISTAKES

- Comparing a managed service's price with VM prices only, ignoring people.
- Forgetting data transfer, which can exceed compute for media and analytics.
- Sizing GPUs for peak and running them 24/7 at 20% utilisation.
- Giving precise-looking numbers without stating assumptions.

8 Key terms

TCO: total cost of ownership, including people and operations

Managed premium: extra price of a managed service over raw resources

Tokens per second per GPU: throughput used to size LLM serving

Egress: data leaving the cloud or region, billed per GB

Cache hit rate: share of requests served without reaching the origin

Opportunity cost: what engineers could build instead of operating infrastructure

9 Interview questions

Q1. How do you include cost in a design discussion? BASIC

Treat cost as a design dimension: estimate the main drivers (compute, storage, data transfer, GPUs, managed premiums), compare options by total cost of ownership including people's time, and state the trade-offs explicitly.

Q2. What are common hidden costs? BASIC

Egress and cross-region transfer, NAT and load balancer processing, observability ingestion, and the operational time of self-hosted systems.

Q3. Estimate egress for 50 TB of video a month and how to cut it. SCENARIO

About 50,000 GB × $0.08 ≈ $4,000; a CDN with a high hit rate cuts origin egress sharply and, with volume or committed pricing, lowers the per-GB cost.

Q4. How many GPUs for 40 requests/s at 500 tokens each? SCENARIO

20,000 tokens/s divided by measured throughput per GPU (say 2,500 tokens/s with batching) gives 8, about 10 with headroom; then look at smaller models, caching and batching to reduce it.

Q5. Managed or self-hosted Kafka for a five-person team? SCENARIO

Managed: once you count a third of an engineer for operations, its total cost is lower, and the team stays on the product; revisit at much larger scale.

Q6. Why does an active-active second region more than double some costs? TRICKY

Besides duplicated infrastructure, every replicated write crosses regions and is billed per GB, and you need capacity in each region to absorb the other's traffic on failover.

Q7. Why can a cheaper architecture be the more expensive choice? TRICKY

Lower cloud spend can mean more engineering time, slower delivery, or higher incident risk; TCO and opportunity cost, not the invoice alone, decide.

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